L2 Processing at Syntax–Prosody Interface: A Time-Frequency Analysis

Prior research indicates that, in second language (L2) processing, prosodic boundary cues can support syntactic processing, as in first language (L1) processing. However, the details of this syntax–prosody interface remain unclear, particularly whether L2 learners rely solely on boundary cues shared with their L1 or also draw on L2-specific cues.

Against this background, the present study investigated whether native Cantonese speakers learning English can make use of phrasal stress, an English-specific boundary cue absent in Cantonese, to facilitate syntactic parsing. To contrast the processing of this L2-specific cue with a cue available in both languages, we also examined pause, a boundary cue shared between Cantonese and English. We assessed prosodic boundary cue utilization through alpha (8–12 Hz) and low-beta (15–20 Hz) power modulation, oscillatory markers associated with successful syntactic unification (e.g., Lam et al., 2016). By comparing these responses to prosodic cues that aligned with versus diverged from major syntactic boundaries, we evaluated whether learners effectively used these cues during sentence parsing. To rule out the possibility that non-target-like integration arose from insufficient perceptual sensitivity, we also measured the Closure Positive Shift (CPS), an ERP component indexing prosodic boundary perception (Steinhauer et al., 1999), for both pause and phrasal stress.

The study compared native English speakers (n = 30), advanced Cantonese learners of English (n = 30), and exceptional learners (n = 30). English proficiency was indexed by learners' performance on the English subject of the Hong Kong university entrance examination, with the two learner groups corresponding to the top 4–10% and 0–0.9% of scores, respectively. Participants completed two EEG sessions of auditory sentence comprehension (Pause and Phrasal Stress sessions), separated by an interval of at least one week. Each session comprised three conditions: Congruent 1, Congruent 2, and Incongruent (see Table 1). The primary contrast of interest was between the Congruent 1 and Incongruent conditions, which were segmentally identical but differed in whether prosodic boundary cues aligned with or diverged from major syntactic boundaries. The Incongruent condition included five counterbalanced versions that varied in the location of the misaligned prosodic cue. For each Congruent 1 sentence, only one of these five Incongruent variants was created. Comparing alpha and low-beta power across these two conditions allowed us to assess prosodic boundary integration. Congruent 2 served to index prosodic boundary perception through the CPS effect; the two Congruent conditions were identical up to the critical region except for whether the critical noun ended in plural *–s* (Congruent 1) or possessive *'s* (Congruent 2). As a result, both sentences were fully well-formed prosodically, but only the Congruent 2 sentence exhibited a prosodic boundary at the critical region (R6). Accordingly, greater CPS amplitudes in the Congruent 2 condition relative to Congruent 1 indicated sensitivity to prosodic boundaries.

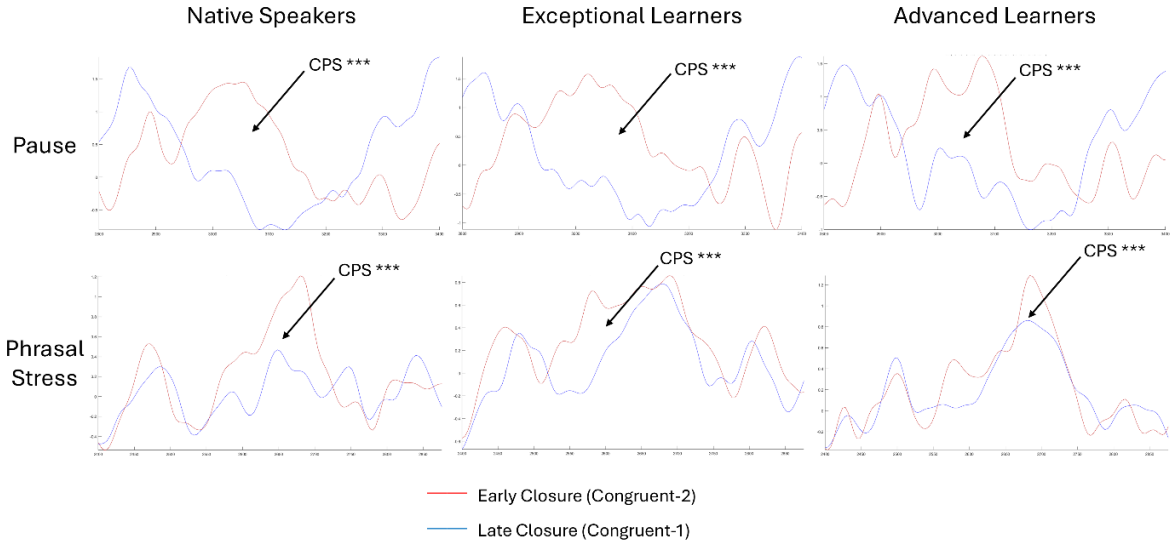
Our mixed-effects models and subsequent pairwise comparisons for the ERP data showed that the CPS effect was elicited in all groups across both sessions (see Figure 1), indicating robust perceptual sensitivity to both pause and phrasal stress boundaries regardless of proficiency level ($ps < .001$). In addition, the mixed-effects models and pairwise comparisons for time-frequency analysis revealed that both native speakers and exceptional learners exhibited higher alpha power for prosody–syntax misalignment than for alignment across both cue types—pause and phrasal stress ($ps < .001$) (see Figure 2), indicating increased processing load under misalignment. In contrast, advanced learners failed to show this significant alpha power increase, despite producing clear CPS effects for both boundary cues. Moreover, in the low-beta band, a corresponding increase in power under prosody–syntax misalignment was observed for exceptional learners, although no such increase was found for native speakers. Taken together, our results suggest that using prosodic boundary cues for syntactic parsing is highly challenging, even for those shared between L1 and L2, and that this ability is observed only in exceptional L2 learners.

Table 1. *Experimental design*

	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11
Pause Condition											
Congruent 1	The	maid #	checked	that	the	kid's	dog #	played	in	the	yard #
Congruent 2	The	maid #	checked	that	the	kids #	played	in	the	yard #	
Incongruent	The #	maid	checked	that #	the	kid's	dog	played	in	the	yard #
	The #	maid	checked	that	the #	kid's	dog	played	in	the	yard #
	The #	maid	checked	that	the	kid's	dog	played	in #	the	yard #
	The	maid	checked	that #	the	kid's	dog	played	in #	the	yard #
	The	maid	checked	that	the #	kid's	dog	played	in #	the	yard #
Phrasal Stress Condition											
Congruent 1	The	**maid**	checked	that	the	kid's	**dog**	played	in	the	**yard**
Congruent 2	The	**maid**	checked	that	the	**kids**	played	in	the	**yard**	
Incongruent	**The**	maid	checked	**that**	the	kid's	dog	played	in	the	**yard**
	The	maid	checked	that	**the**	kid's	dog	played	in	the	**yard**
	The	maid	checked	that	the	kid's	dog	played	**in**	the	**yard**
	The	maid	checked	**that**	the	kid's	dog	played	**in**	the	**yard**
	The	maid	checked	that	**the**	kid's	dog	played	**in**	the	**yard**

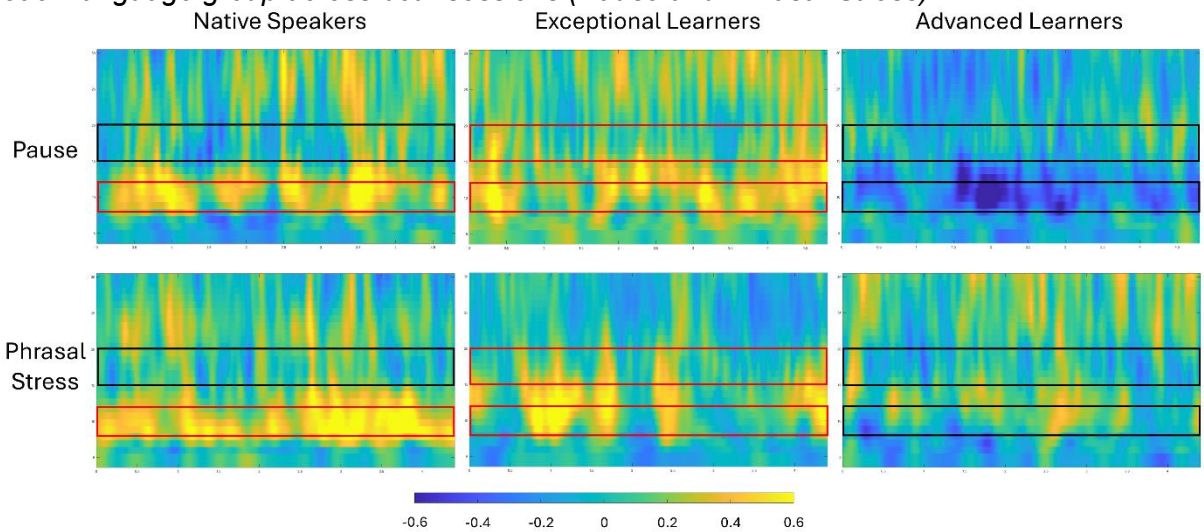
Note. # indicates a pause insertion. Words in bold were superimposed with phrasal stress. The shaded region (Region 6) denotes the critical area for the CPS analysis.

Figure 1. *CPS waveforms across experimental conditions for each language group*



Note. *** indicates $p < .001$.

Figure 2. *Alpha and low-beta power difference plots (Incongruent minus Congruent 1) for each language group across both sessions (Pause and Phrasal-Stress)*



Note. The upper and lower boxes indicate the low-beta (15–20 Hz) and alpha ranges (8–12 Hz), respectively. Red boxes indicate a significant power increase ($p < .01$).

References: Lam, Schoffelen, Uddén, Hultén, & Hagoort (2016). Neural activity during sentence processing as reflected in theta, alpha, beta, and gamma oscillations. *NeuroImage*; Steinhauer, Alter, & Friederici (1999). Brain potentials indicate immediate use of prosodic cues in natural speech processing. *Nature neuroscience*.